

2 (**short**) A two-dimensional air flow has circular stream lines, as shown in Fig. 2. The velocity distribution is given by $V(r) = c/r$, where c is a constant and r is the radial distance. At radius $r_0 = 0.2$ m the velocity is $V_0 = 20.0$ m s⁻¹. Assume air to be incompressible with density $\rho = 1.2$ kg m⁻³.

(a) Calculate the pressure difference between the streamlines at $r = r_0$ and $r = r_1 = 0.1$ m. [6]

(b) Prove whether or not Bernoulli's equation can be applied between the streamlines at different radii. [4]

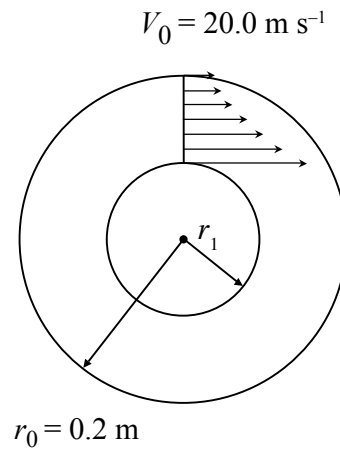


Fig. 2